

Questions			Answers	Marks
2	(a)		molecules has <u>component of velocity in 3 directions</u>, hence $c^2 = c_x^2 + c_y^2 + c_z^2 \text{ ----- (1)}$	M1
			their motions are random, so averaging gives $\langle c_x^2 \rangle = \langle c_y^2 \rangle = \langle c_z^2 \rangle \text{ ----- (2)}$	M1
			Hence substitute the result of (2) into (1), then $\langle c^2 \rangle = 3\langle c_x^2 \rangle$	A1

			Hence for N molecules, we have $pV = \frac{1}{3}Nm\langle c^2 \rangle$	A0
	(b)	(i)	pressure $pV = nRT$ or $p = \frac{nRT}{L^3} = \frac{3 \times 8.31 \times (20.0 + 273.15)}{0.200^3}$	M1
			so force on each sides $F = pA = \frac{3 \times 8.31 \times (20.0 + 273.15)}{0.200^3} \times 0.200^2$	M1
			$F = 36\,500\text{ N}$	A1
		(ii)	particles would <u>attract</u> one another	B1
			so average force on wall will <u>decrease</u>	B1

2	(a)	(i)	$p = \frac{1}{3} \frac{Nm\langle c^2 \rangle}{V}$ [M1]
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